

Task 1 – CSV File: Student Marks Record

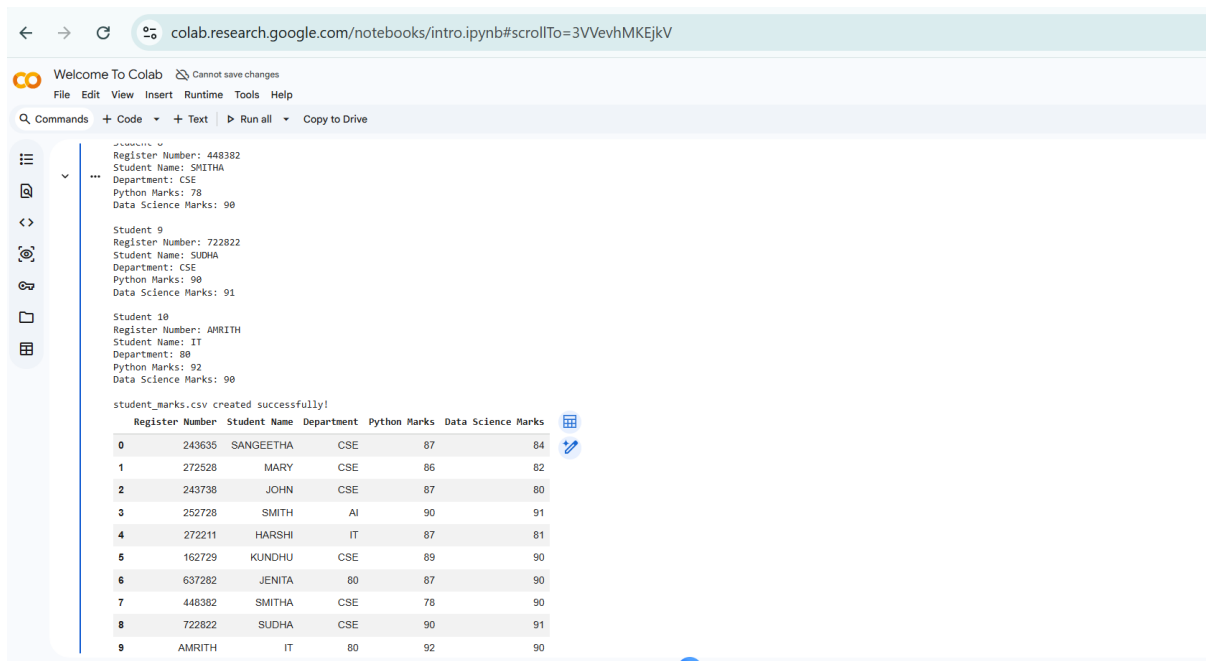
Problem Statement

Develop a Python program to collect the details of 10 students and store them in a CSV file named student_marks.csv.

The file should contain:

- Register Number
- Student Name
- Department
- Python Marks
- Data Science M

Output:



The screenshot shows a Google Colab notebook interface. The top bar displays the URL: `colab.research.google.com/notebooks/intro.ipynb#scrollTo=3VVevhMKEjKv`. The notebook content includes three code cells for individual students and a final cell that displays a CSV file named `student_marks.csv`.

Code Cell 1 (Student 8):

```
Register Number: 448382
Student Name: SMITHA
Department: CSE
Python Marks: 78
Data Science Marks: 90
```

Code Cell 2 (Student 9):

```
Student 9
Register Number: 722822
Student Name: SUDHA
Department: CSE
Python Marks: 90
Data Science Marks: 91
```

Code Cell 3 (Student 10):

```
Student 10
Register Number: AMRITH
Student Name: IT
Department: 80
Python Marks: 92
Data Science Marks: 90
```

Code Cell 4 (CSV File):

```
student_marks.csv created successfully!
```

	Register Number	Student Name	Department	Python Marks	Data Science Marks
0	243635	SANGEETHA	CSE	87	84
1	272528	MARY	CSE	86	82
2	243738	JOHN	CSE	87	80
3	252728	SMITH	AI	90	91
4	272211	HARSHI	IT	87	81
5	162729	KUNDHU	CSE	89	90
6	637282	JENITA	80	87	90
7	448382	SMITHA	CSE	78	90
8	722822	SUDHA	CSE	90	91
9	AMRITH	IT	80	92	90

Task 2 – CSV File: Employee Database

Problem Statement

Develop a Python program to collect employee information and store it in a CSV file named employees.csv.

The file should contain:

- Employee ID
- Employee Name

- Department
- Designation
- Salary

Store details of at least 5 employees.

Output:

The screenshot shows a Google Colab interface. The main code cell contains the following text:

```
Employee 3
Employee ID: 345
Employee Name: SMITH
Department: AI
Designation: MANAGER
Salary: 100000

Employee 4
Employee ID: 235
Employee Name: SANGEETHA
Department: IT
Designation: SOFTWARE DEVELOPER
Salary: 80000

Employee 5
Employee ID: 276
Employee Name: KUNDHU
Department: AI
Designation: TEAM LEAD
Salary: 95000

employees.csv created successfully!
```

Below the code, a table icon indicates a CSV file named 'employees.csv' has been created. The table contains the following data:

	Employee ID	Employee Name	Department	Designation	Salary
0	243	SUNAINA	IT	SOFTWARE DEVELOPER	80000.0
1	231	LOKESH	IT	SOFTWARE DEVELOPER	100000.0
2	345	SMITH	AI	MANAGER	100000.0
3	235	SANGEETHA	IT	SOFTWARE DEVELOPER	80000.0
4	276	KUNDHU	AI	TEAM LEAD	95000.0

Task 3 – JSON File: Faculty Information System

Problem Statement

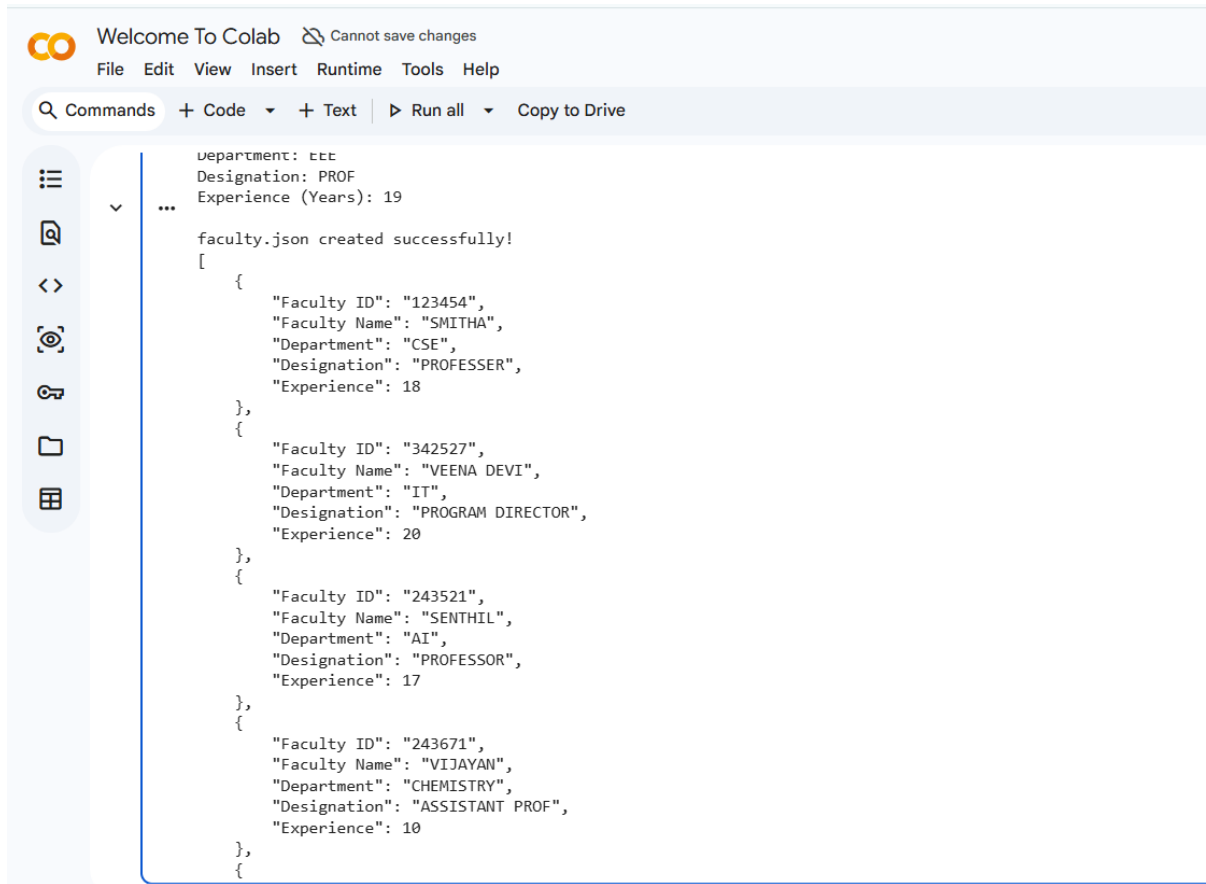
Create a Python program to collect faculty details and store them in a JSON file named faculty.json.

Each faculty record should contain:

- Faculty ID
- Faculty Name
- Department
- Designation
- Experience (Years)

Store details of at least 5 faculty members.

Output:



The screenshot shows the Google Colab interface. At the top, it says "Welcome To Colab" and "Cannot save changes". Below the menu bar (File, Edit, View, Insert, Runtime, Tools, Help), there is a toolbar with "Commands", "+ Code", "+ Text", "Run all", and "Copy to Drive". The left sidebar contains icons for file management. The main output area displays the following text:

```
Department: EEE
Designation: PROF
Experience (Years): 19
...
faculty.json created successfully!
[
  {
    "Faculty ID": "123454",
    "Faculty Name": "SMITHA",
    "Department": "CSE",
    "Designation": "PROFESSOR",
    "Experience": 18
  },
  {
    "Faculty ID": "342527",
    "Faculty Name": "VEENA DEVI",
    "Department": "IT",
    "Designation": "PROGRAM DIRECTOR",
    "Experience": 20
  },
  {
    "Faculty ID": "243521",
    "Faculty Name": "SENTHIL",
    "Department": "AI",
    "Designation": "PROFESSOR",
    "Experience": 17
  },
  {
    "Faculty ID": "243671",
    "Faculty Name": "VIJAYAN",
    "Department": "CHEMISTRY",
    "Designation": "ASSISTANT PROF",
    "Experience": 10
  },
  {
```

Task 4 – JSON File: Product Inventory Management

Problem Statement

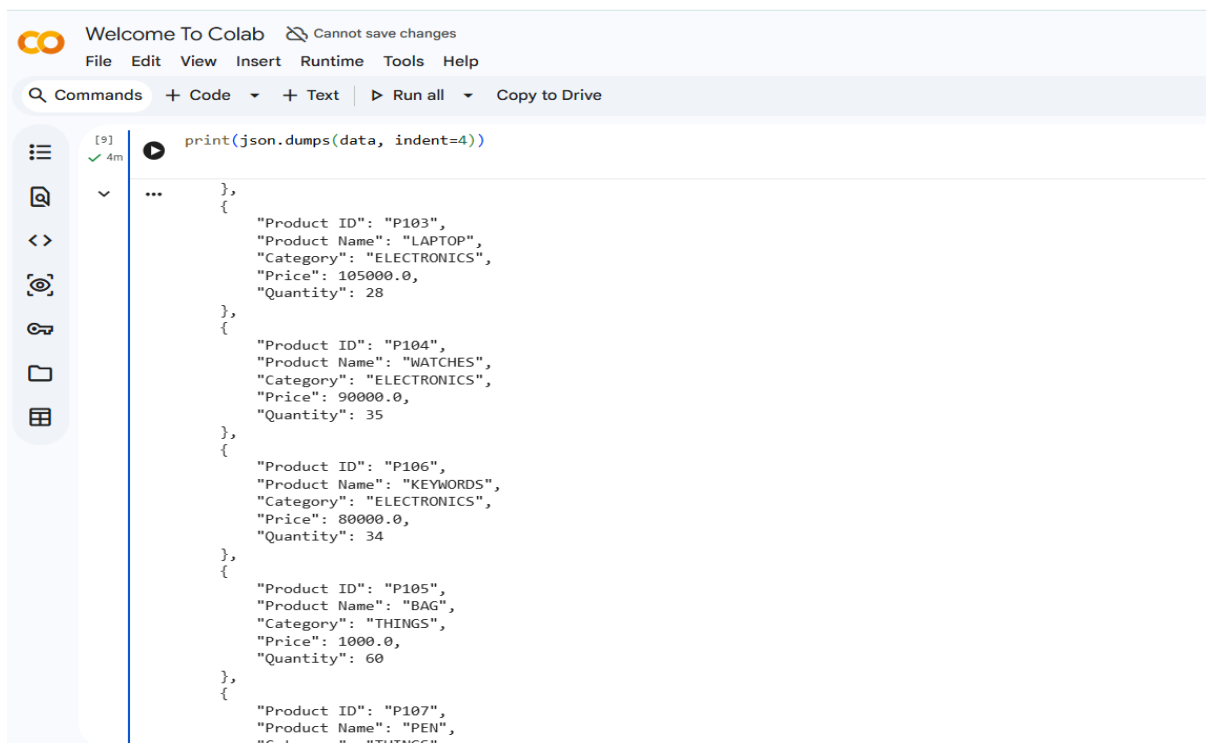
Develop a Python program to collect product information and store it in a JSON file named products.json.

Each product should contain:

- Product ID
- Product Name
- Category
- Price
- Quantity Available

Store details of at least 10 products.

Output:



The screenshot shows a Google Colab interface. At the top, there's a header with the Colab logo, 'Welcome To Colab', and a warning 'Cannot save changes'. Below this is a menu bar with 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help'. A toolbar contains 'Commands', '+ Code', '+ Text', 'Run all', and 'Copy to Drive'. The main area shows a code cell with the following Python code:

```
print(json.dumps(data, indent=4))
```

The code cell has a status bar showing '[9]', a green checkmark, and '4m'. The output of the code is a JSON array of product objects, formatted with an indent of 4 spaces:

```
[
  {
    "Product ID": "P103",
    "Product Name": "LAPTOP",
    "Category": "ELECTRONICS",
    "Price": 105000.0,
    "Quantity": 28
  },
  {
    "Product ID": "P104",
    "Product Name": "WATCHES",
    "Category": "ELECTRONICS",
    "Price": 90000.0,
    "Quantity": 35
  },
  {
    "Product ID": "P106",
    "Product Name": "KEYWORDS",
    "Category": "ELECTRONICS",
    "Price": 80000.0,
    "Quantity": 34
  },
  {
    "Product ID": "P105",
    "Product Name": "BAG",
    "Category": "THINGS",
    "Price": 1000.0,
    "Quantity": 60
  },
  {
    "Product ID": "P107",
    "Product Name": "PEN",
    "Category": "THINGS",
    "Price": 1000.0,
    "Quantity": 60
  }
]
```

Task 5 – XML File: Course Information Database

Problem Statement

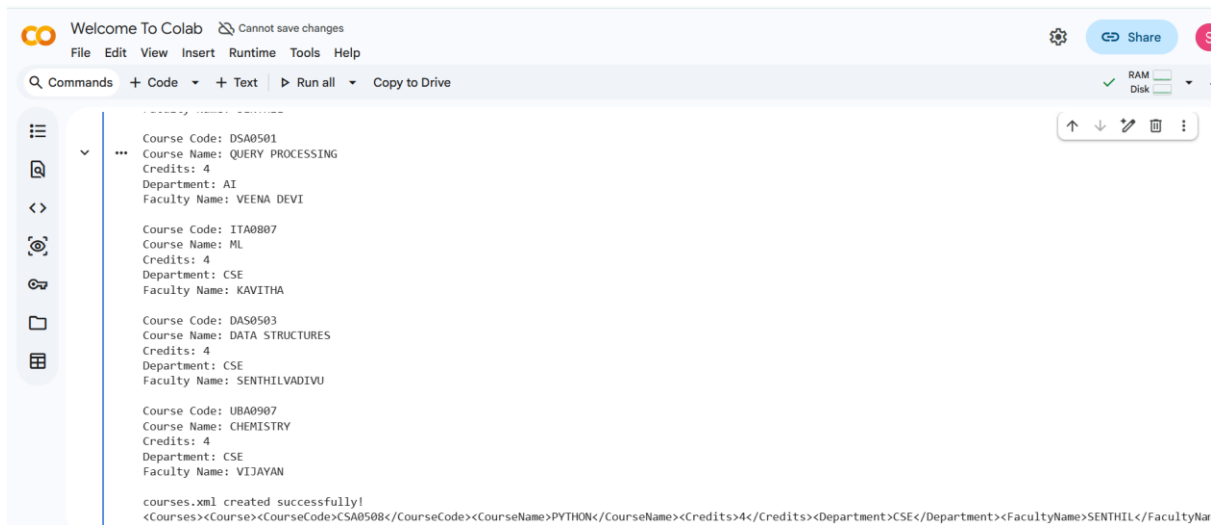
Create a Python program to collect course details and store them in an XML file named courses.xml.

Each course should contain:

- Course Code
- Course Name
- Credits
- Department
- Faculty Name

Store details of at least 5 courses.

Output:



The screenshot shows a Google Colab notebook interface. The top bar includes the Colab logo, 'Welcome To Colab', and a 'Cannot save changes' warning. Below this is a menu bar with 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help'. A toolbar contains 'Commands', '+ Code', '+ Text', 'Run all', and 'Copy to Drive'. On the right, there's a 'Share' button and a RAM/Disk usage indicator. The left sidebar shows icons for file explorer, search, code editor, runtime, and help. The main area displays the following text:

```
Course Code: DSA0501
Course Name: QUERY PROCESSING
Credits: 4
Department: AI
Faculty Name: VEENA DEVI

Course Code: ITA0807
Course Name: ML
Credits: 4
Department: CSE
Faculty Name: KAVITHA

Course Code: DAS0503
Course Name: DATA STRUCTURES
Credits: 4
Department: CSE
Faculty Name: SENTHILVADIVU

Course Code: UBA0907
Course Name: CHEMISTRY
Credits: 4
Department: CSE
Faculty Name: VIJAYAN

courses.xml created successfully!
<Courses><Course><CourseCode>CSA0508</CourseCode><CourseName>PYTHON</CourseName><Credits>4</Credits><Department>CSE</Department><FacultyName>SENTHIL</FacultyName>
```

Task 6 – XML File: Hospital Patient Management System

Problem Statement

Develop a Python program to collect patient information and store it in an XML file named patients.xml.

Each patient record should contain:

- Patient ID
- Patient Name
- Age
- Disease
- Doctor Assigned

Store details of at least 10 paths.

Output:



The screenshot shows the Google Colab interface. At the top, there's a header with the Colab logo, 'Welcome To Colab', and a warning 'Cannot save changes'. Below this is a menu bar with 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help'. A toolbar contains 'Commands', '+ Code', '+ Text', 'Run all', and 'Copy to Drive'. On the right, there's a 'Share' button and RAM/Disk usage indicators. A left sidebar has icons for file explorer, search, code editor, terminal, and help. The main area displays patient data and XML output.

```
Patient ID: P464
Patient Name: SMITHA
Age: 45
Disease: BP
Doctor Assigned: RAHUL

Patient ID: P937
Patient Name: VIJAI
Age: 35
Disease: INCREASE IN SUGAR LEVELS
Doctor Assigned: MARY

Patient ID: P363
Patient Name: RAHUL
Age: 32
Disease: BP
Doctor Assigned: RAHUL

Patient ID: P373
Patient Name: MARY
Age: 32
Disease: BP
Doctor Assigned: RAHUL

patients.xml created successfully!
<Patients><Patient><PatientID>P111</PatientID><PatientName>SAMPATH</PatientName><Age>26</Age><Disease>BP</Disease><DoctorAssigned>RAHUL</DoctorAssigned></Pati
```